

Risk Management and Valuation

Group Coursework 1A (AIFS)

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Executive Summary

This report analyses hedging strategies for the AIFS to protect earnings against significant currency exposure (EUR/GBP costs vs. USD revenue) and participant volume uncertainty. As AIFS guarantees its prices a year in advance, the company faces potential losses if actual exchange rate and sales volume deviates from forecasts and no hedging is in place.

Our analysis finds that pure hedging strategies are suboptimal. Forward contracts lock in rates to prevent unfavourable movements. However, given AIFS's volume uncertainty, they create significant risk, potentially forcing AIFS to purchase excess currency, or currency at unfavourable rates (**Appendix 4a**). In contrast, currency options provide flexibility and cap maximum costs, but they incur high premiums (5% of notional value), making them costly as a sole solution.

Therefore, we recommend a mixed strategy that balances cost and flexibility. Stress-test modelling (**Appendix 4d**) identifies the optimal strategies for the divisions: 40% forwards and 60% options for ACIS, which guarantees an EBIT floor of \$0.385M, even if the number of participants drops from 25,000 to 10,000, and 30% forwards and 70% options for College (per policy constraints).

Q1: Should AIFS not hedge its FX exposure?

In a frictionless world, a firm would have no incentive to hedge its foreign exchange exposure. From a standard corporate finance perspective, hedging does not create value because it can simply reallocate cash flows across different states of the world without changing their expected value. In addition, hedging entails transaction costs, such as option premiums, which will reduce expected value.

However, AIFS operates in a real-world environment with some key risks that generate frictions:

First, the firm is exposed to substantial bottom-line risk arising from exchange rate movements. Since two of AIFS's major divisions served American students travelling abroad, AIFS's revenues are mostly gained in US dollars, while a large share of its costs is in euros and British pounds. If the dollar depreciated sharply, the resulting increase in the cost base could be large enough to threaten the firm's operations. As illustrated in Appendix 1, remaining unhedged exposes AIFS to substantial cost variability driven by exchange rate movements.

Second, AIFS faces significant volume risk, particularly in its High School Travel (ACIS) division. Sales volumes are highly sensitive to geopolitical and macroeconomic shocks such

as wars, terrorist attacks, or political instability. Historical experience shows that enrolments can fall by as much as 60 percent of following such events. Since foreign currency is typically purchased based on projected rather than realised sales volumes, the firm may end up with excess or insufficient foreign currency exposure, which brings uncertainty. What's more, volume risk sometimes occurs with currency risk.

Third is the competitive pricing. AIFS's price guarantee is a core element of its value proposition and customer loyalty, but it also prevents the firm from passing cost increases on to customers when exchange rates move unfavourably. Consequently, foreign exchange shocks will directly influence AIFS's operating profits and cash flows without much buffering. As a result, low-cash-flow states are especially costly for the firm. Losses in such states can trigger financial distress and limit the firm's ability to apply financial operations or pursue future opportunities.

Therefore, the objective of risk management for AIFS is not simply to stabilise costs or reduce earnings volatility. Rather, the goal is to protect cash flows in adverse states of the world, when cash is not enough to support AIFS's operating needs. In the presence of these frictions, hedging can create value by redistributing cash flows away from the worst outcomes and reducing the likelihood or severity of financial distress. Although hedging does not create value in theory, under real-world frictions, AIFS should hedge its FX exposure to protect cash flows in adverse states.

Q2: Should AIFS hedge with forward contracts?

AIFS should hedge with the forward contract due to its foreign currency risk. Forward contracts enable AIFS to eliminate exchange rate risk by fixing in advance the exchange rate at which future foreign currency payments will be made. This is particularly relevant for the College division, where program prices are set approximately one year before students enroll and euro costs are dominated. Without hedging, the dollar cost of these euro expenses would depend on the spot exchange rate prevailing at the time of payment, exposing AIFS to potentially adverse currency movements.

2.1 Forward Contracts as a Tool for Hedging Exchange Rate Risk

By entering into a forward contract at the time prices are set, AIFS commits to purchasing a predetermined amount of euros at a fixed forward exchange rate on a maturity date. The forward contract is agreed at the stage, but no cash is exchanged at first. At maturity, when the actual costs are incurred, AIFS settles the contract and exchanges dollars for euros at the

agreed forward rate, regardless of the future spot rate. As a result, the dollar value of future costs becomes known with certainty at the time of pricing.

Let S_0 denote the spot FX quote in \$/£ (USD per GBP), $r_{dollars}$ the USD 1-year interest rate, and r_{pounds} the GBP 1-year interest rate. The 1-year forward rate $F_{0,1}$ satisfies

$$F_{0,1} = S_0 \frac{1 + r_{dollars}}{1 + r_{pounds}}.$$

Using the supporting slide inputs $S_0 = 1.57$, $r_{dollars} = 1.55\%$, $r_{pounds} = 2.35\%$,

$$F_{0,1} = 1.57 \times \frac{1.0155}{1.0235} \approx 1.558.$$

Hence, the bank can quote a lock-in forward rate consistent with no-arbitrage pricing.

Let V be the realised number of participants and assume euro-denominated variable cost per participant is $c_{eur} = 1000$.

Let S be the realised spot exchange rate in \$/EUR. Total euro cost is

$$C_{eur}(V) = c_{eur}V.$$

If AIFS is unhedged, total dollar cost is linear in S :

$$C_{dollars}^{\text{unhedged}}(S, V) = S c_{eur}V.$$

For the base forecast $V_0 = 25,000$, $C_{eur} = 25\text{m } eur$, so at the three FX scenarios:

$$C_{dollars}^{\text{unhedged}}(1.01, 25k) = 25.25\text{m}, \quad C_{dollars}^{\text{unhedged}}(1.22, 25k) = 30.50\text{m}, \quad C_{dollars}^{\text{unhedged}}(1.48, 25k) = 37.00\text{m}.$$

Hence FX risk creates a cost range of \$11.75m.

A 100% forward hedge at $F = 1.22$ locks the dollar cost of the forecast euro amount:

$$C_{dollars}^{\text{fwd}}(V_0) = F \cdot c_{eur}V_0 = 1.22 \times 25\text{m} = \$30.5\text{m},$$

which is independent of S (flat line in the “Hedging with Forwards” figure in Appendix 2a).

2.2 Hedging Cash Flows Rather Than Stabilising Costs

The objective of AIFS's hedging policy is not just to stabilise operating costs, but to protect the firm's overall cash flows against adverse exchange rate movements. Focusing solely on cost stabilisation can be misleading, as costs represent only one component of the firm's total cash flow. From a financial perspective, the relevant risk is the possibility that exchange rate fluctuations reduce net operating cash flows to a level insufficient to meet the firm's cash needs, such as funding ongoing operations, servicing financial obligations, or financing positive NPV investment opportunities.

Assume revenues net of non-travel costs are \$1350 per participant (as in the slide). Then operating cash flow (unhedged) is

$$\Pi^{\text{unhedged}}(S, V) = V(1350 - 1000S).$$

Thus FX risk directly affects operating cash flows and the probability that cash flows fall short of cash needs (e.g. internal funding for investment, servicing obligations, avoiding distress), rather than merely the volatility of the individual cost project.

2.3 Volume Risk and the Limitations of Forward Contracts

While forward contracts are effective in eliminating exchange rate risk, they do not address volume risk arising from uncertainty about future enrollment amounts. At the time hedging decisions are made, AIFS does not yet know how many students will enroll in its programs in the next year. As a result, the firm must hedge based on projected, rather than realised, foreign currency requirements.

As in the supporting slides, suppose AIFS hedges the *forecast* euro requirement based on $V_0 = 25,000$ via a forward to buy

$$N_{eur} = c_{eur} V_0 = 25\text{m} \quad \text{at} \quad F = 1.22.$$

At maturity, realised volume V may differ from V_0 .

- **High volume scenario** ($V > V_0$): **under-hedged**. AIFS must buy the extra euros $c_{eur}(V - V_0)$ at spot S . Total dollar cost becomes

$$C_{dollars}^{\text{fwd}}(S, V) = c_{eur}(FV_0 + S(V - V_0)), \quad V \geq V_0.$$

Hence operating cash flow under the forward hedge is

$$\Pi^{\text{fwd}}(S, V) = 1350V - 1000 (FV_0 + S(V - V_0)), \quad V \geq V_0,$$

which retains some exposure to a weakening dollar (higher S) on the *incremental* volume only. For the slide example $V = 30,000$ (30m total cost),

$$C_{\text{dollars}}^{\text{fwd}}(S, 30k) = 1000 (1.22 \times 25,000 + S \times 5,000) = 30.5\text{m} + 5\text{m} \cdot S,$$

- **Crisis scenario ($V < V_0$): over-hedged.** AIFS is committed to buying too many euros via the forward. It buys $c_{\text{eur}}V_0$ euros at F , uses $c_{\text{eur}}V$ euros for actual costs, and must sell the surplus $c_{\text{eur}}(V_0 - V)$ euros at spot S . The net dollar cost is

$$C_{\text{dollars}}^{\text{fwd}}(S, V) = c (FV_0 - S(V_0 - V)), \quad V \leq V_0.$$

Therefore operating cash flow is

$$\Pi^{\text{fwd}}(S, V) = 1350V - 1000 (FV_0 - S(V_0 - V)), \quad V \leq V_0.$$

For the slide crisis example $V = 10,000$ (need 10m but hedged 25m),

$$C_{\text{dollars}}^{\text{fwd}}(S, 10k) = 1000 (1.22 \times 25,000 - S \times 15,000) = 30.5\text{m} - 15\text{m} \cdot S,$$

In particular, if the dollar strengthens (S falls), the sale proceeds on the surplus euros shrink, creating *additional losses* precisely when volume (and revenues) are already low.

To mitigate the downside risk associated with over-hedging, AIFS should complement forward contracts with currency options. Unlike forwards, options provide the right, but not the obligation, to exchange currency at a predetermined rate. At maturity, AIFS can choose whether to exercise the option based on the prevailing spot exchange rate. If the euro depreciates and enrollment turns out to be lower than expected, the firm can allow the option to expire and purchase only the required amount of foreign currency at the more favorable spot rate.

Q3: Should AIFS hedge with option contracts?

3.1 From price hedging to downside cash-flow protection

By locking in a future purchase price, forward contracts eliminate exchange-rate uncertainty, but they also impose a commitment: regardless of actual program volume, AIFS is required

to acquire a particular amount of foreign currency at maturity. This is significant since AIFS is subject to both volume risk brought on by geopolitical shocks and foreign exchange risk. The marginal value of liquidity is largest in adverse crisis states where enrollments plummet, but if AIFS gets over-hedged in those states, a forward hedge may result in hedging losses. Options are attractive in this case because they provide state-contingent protection, which eliminates the need to trade in the event that the underlying exposure does not materialize while guaranteeing against a decreasing dollar. This makes safeguarding downside operating cash flows in circumstances where financial distress is most likely the goal of hedging instead of minimizing cost volatility.

3.2 Option hedge design and payoff: a cost cap with upside preservation

Consider AIFS's base euro exposure from the supporting slides: total euro costs of

$$Q = 25 \text{ million euros.}$$

To hedge against a weakening dollar (a higher spot rate S measured in USD per EUR), AIFS can buy EUR call options (equivalently, USD put options) with strike (K) and premium (p):

$$K = 1.22 \text{ (\$/EUR)}, p = 0.061 \text{ (\$/EUR)},$$

so total premium is $pQ = 0.061 \times 25 = 1.525$ million USD.

Unhedged dollar cost. If AIFS remains unhedged, its dollar cost is linear in the exchange rate:

$$C_{\text{unh}}(S) = QS. \tag{1}$$

Forward-hedged dollar cost. A full forward hedge locks in cost at QK :

$$C_{\text{fwd}}(S) = QK. \tag{2}$$

Option-hedged dollar cost (full notional). With EUR calls, AIFS pays the premium up front and at maturity chooses the cheaper of (i) buying euros at spot S or (ii) exercising at K :

$$C_{\text{opt}}(S) = Q \min(S, K) + pQ. \tag{3}$$

Equation (3) highlights the key economic feature: options impose a *cap* on the effective exchange rate paid by AIFS. Specifically,

$$C_{\text{opt}}(S) = \begin{cases} QS + pQ, & S \leq K \quad (\text{dollar strong: do not exercise}) \\ QK + pQ, & S > K \quad (\text{dollar weak: exercise and cap cost}). \end{cases}$$

Thus, relative to forwards, options preserve upside (benefiting from a stronger dollar) while still protecting against the left-tail outcome (a weaker dollar). Relative to remaining unhedged, options truncate the right tail of costs by converting linear exposure into a kinked, concave profile. This formulation reflects the base-case notional hedge. Under volume uncertainty, only the required euro amount would be exercised, which becomes crucial in crisis states (discussed in Section 3.4).

3.3 Illustration using the base-case scenarios in the supporting slides

The table in appendix 3 replicates the base-case comparison shown in the supporting slides for three illustrative exchange rates (strong/stable/weak dollar). The main takeaway is that options exchange a known premium for (i) cost protection when S rises above K and (ii) the ability to benefit when S falls below K .

3.4 Why options are especially valuable under volume uncertainty (crisis states)

The case's central complication is that AIFS does not know *ex ante* how many students will enroll, so the “correct” hedge notional is uncertain. In the crisis scenario (e.g. $N = 10,000$ participants), the supporting slides express operating cash flow (or operating profit) as

$$\Pi_{\text{unh}}(S; N) = N(1350 - 1000S), \quad (4)$$

where 1350 USD is revenue net of non-travel costs per participant and 1000 EUR is the EUR-denominated travel cost per participant. Equation (4) makes clear that a higher S (weaker dollar) directly compresses margins.

The forward-commitment problem. With a forward hedge sized to the base case, AIFS is forced to acquire euros even when they are not needed. If realized volume is low, a full forward hedge becomes an over-hedge; the firm may have to purchase excess euros at K and then effectively unwind at spot S . This can generate hedging losses in crisis states, compounding low

operating profits. In other words, futures can result in a "wrong-way" exposure since hedging losses occur at the same moment when internal cash flows are already poor.

Options remove the obligation channel. Options change the risk structure rather than merely shifting the exchange rate at which euros are acquired. Because the option is a right, not an obligation, AIFS can: (i) exercise only when $S > K$ (protecting against a weakening dollar), and (ii) avoid acquiring excess euros when volume is below expectations. This flexibility is especially useful during times of crisis, when maintaining a minimal level of cash flow is more crucial than lowering cost variance during regular times.

3.5 Managerial conclusion and implication for Q4

AIFS should hedge with option contracts because options provide downside protection against a weakening dollar while mitigating the commitment risk that makes forward hedges fragile under volume uncertainty. The trade-off is the premium: options reduce expected cash flows in benign states by the upfront cost pQ . Therefore, options are best interpreted as disaster insurance that improves survivability in adverse states. This naturally motivates the next question: how does a firm combine forwards (low cost, high commitment) and options (high cost, high flexibility) to achieve robust protection at lower premium expense?

Q4: Should AIFS use a combination of options and forwards?

4.1 Cost minimization perspective

AIFS should use both currency options and forward contracts to hedge against FX risks. From a cost perspective in FX hedging, the usage of options puts a cap on costs if the USD depreciates, while allowing the firm to capture lower costs if the USD appreciates, but the price to pay is the option premium. On the other hand, while the forward contract imposes no costs, the firm has to lock in a fixed rate now for cash payments later, which can be unfavourable if the USD appreciates in future.

The AIFS shifting box summarised by Archer-Lock (**Appendix 4a**) demonstrates that no single hedging instrument dominates in all scenarios.

- **Square 1 (Crisis / Low Volume):** This scenario implies a currency surplus. If AIFS is locked into a surplus forward, it would crystallise a loss. Therefore, options are essential here to provide flexibility.

- **Squares 2 and 4 (Favourable Rate / Volume Mismatch):** The exchange rate is in-the-money (ITM). Both forwards and options provide a gain, but the gain is larger with forward contracts as they do not incur an upfront premium
- **Square 3 (Unfavourable Rate):** The exchange rate is out-of-the-money (OTM). Options excels here, as AIFS can abandon the contract and purchase currency at the more favourable spot rate.

Therefore, a mixed strategy with a combination of both options and forwards is justified, as it ensures that AIFS is never fully exposed to the worst outcomes of any squares. The best hedging strategy for both the College and ACIS division is thus a 25% option to 75% forward ratio (**Appendix 4b**) - which reproduces the lowest costs while taking advantage of the benefits of both contracts due to their individual mechanics.

4.2 Cash flow protection perspective

While cost is an essential metric for determining an optimal hedging combination, it is more important for firms to seek downside protection, ensuring that their cash flows are sufficient to cover their cash needs during crisis scenarios. Examples of cash needs include interest payments, positive NPV opportunities, and payroll. Without sufficient cash flows, AIFS may not tide through market downturns given its relatively small size of 1000+ employees (source: leadiQ.com). A small-sized firm will have harder access to financing due to credit tightening and lesser diversification to generate other sources of income, given its limited services.

As such, we determine the optimal ratio of options and forwards by computing the combinations of options and forwards and their respective operating profit (revenue - cost) to identify the greatest lower bound on the operating profit (**Appendix 4d**) in a crisis scenario. As seen in Appendix 4e, the 40% forward and 60% options combination provides the best downside protection in the crisis scenario where the minimum operating profit is at US\$0.385M for ACIS. Other combinations (**Appendix 4c**) either have operating profit going negative, close to zero, or reaches zero somewhere along the same exchange rate spectrum which is not ideal as AIFS needs to have a positive cashflow to maintain liquidity in unfavourable market conditions.

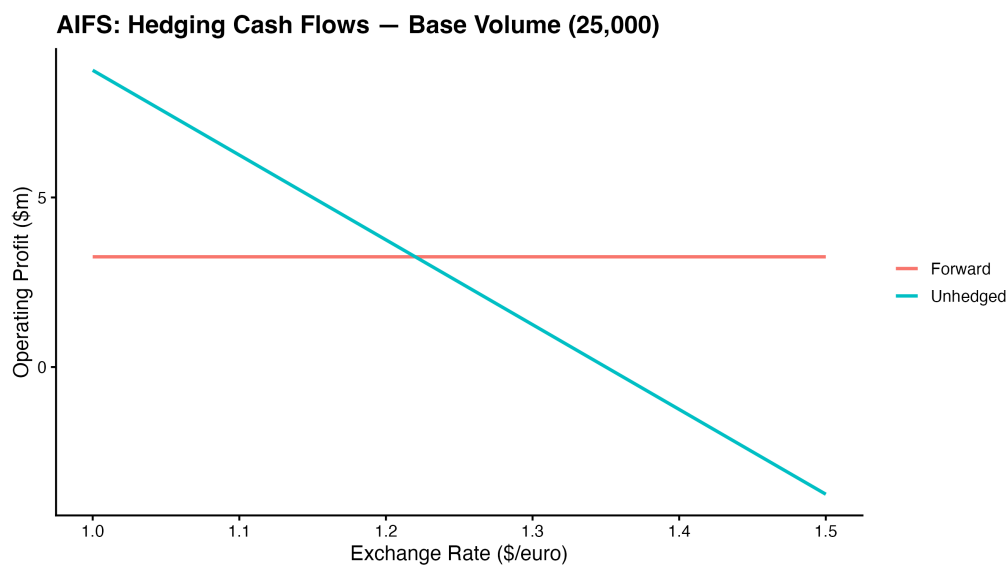
Taking AIFS's internal policy into account, the optimal model of 40% forward and 60% options for ACIS remains viable as it respects the ACIS's option limit of 50%. However, the optimal model exceeds the 30% options cap for the College division. Therefore, to comply with the policy while maximising protection, we adjust the College's division mix to 70% forwards and 30% options.

Appendix

Appendix 1: Table of cost outcomes without hedging

Scenario	USD/EUR spot rate S	Total cost (USD m)	Deviation vs. $S = 1.22$ (USD m)
Strong dollar	1.01	25.25	+5.25
Stable dollar	1.22	30.50	0.00
Weak dollar	1.48	37.00	-6.50

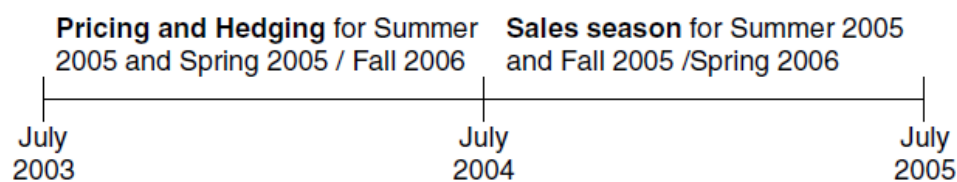
Appendix 2a: Hedging with forward (base case)



Appendix 3: Table of Base-case costs under unhedged, forward, and option hedges

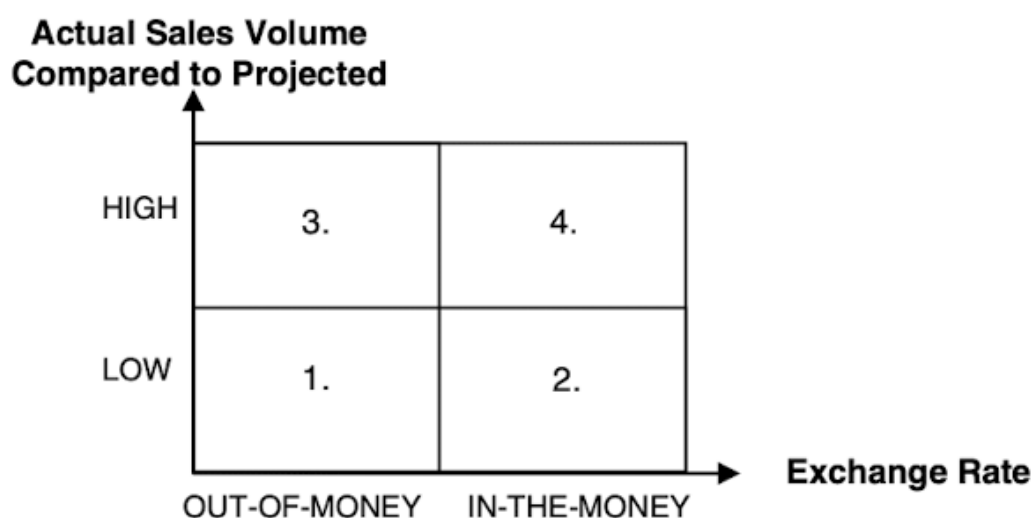
	$S = 1.01$	$S = 1.22$	$S = 1.48$
Unhedged: $C_{\text{unh}}(S) = QS$ (USD m)	25.25	30.50	37.00
Forward: $C_{\text{fwd}}(S) = QK$ (USD m)	30.50	30.50	30.50
Option: $C_{\text{opt}}(S)$ (USD m)	26.775	32.025	32.025

Appendix 3a: Hedging and pricing timeline for college department



Appendix 4a: AIFS Shifting Box

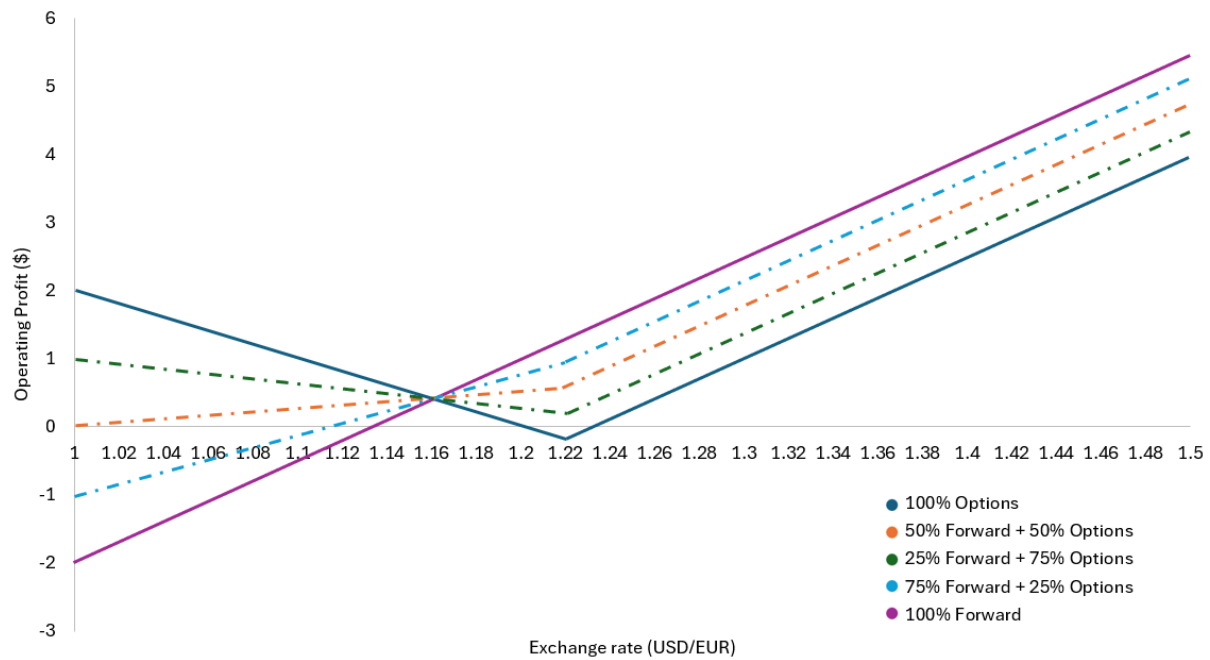
AIFS Shifting Box



Appendix 4b: Calculation of optimal costs (base case)

Projected sales volume (no. of participant)	25,000		
Cost per participant (€)	1,000		
Total cost in euros (€)	25,000,000		
50% options and 50% forward			
Euros needed for options (€)	12,500,000	Euros needed for options (€)	18,750,000
Cost of option (\$1.22 + 5% premium) (\$)	1.281	Cost of option (\$1.22 + 5% premium) (\$)	1.281
Total cost incurred in USD with option (\$)	16,012,500	Total cost incurred in USD with option (\$)	24,018,750
Euros needed for forward (€)	12,500,000	Euros needed for forward (€)	6,250,000
Rate locked with forward contract (\$/€)	1.22	Rate locked with forward contract (\$/€)	1.22
Total cost incurred in USD with forward contract (\$)	15,250,000	Total cost incurred in USD with forward contract (\$)	7,625,000
Total cost of combination (\$)	31,262,500	Total cost of combination (\$)	31,643,750
25% options and 75% forward			
Euros needed for options (€)	6,250,000		
Cost of option (\$1.22 + 5% premium) (\$)	1.281		
Total cost incurred in USD with option (\$)	8,006,250		
Euros needed for forward (€)	18,750,000		
Rate locked with forward contract (\$/€)	1.22		
Total cost incurred in USD with forward contract (\$)	22,875,000		
Total cost of combination (\$)	30,881,250		

Appendix 4c: Graph of combinations of options and forward (crisis case)



From this figure I have plotted in Excel, one can see that a 75% forward + 25% option strategy has a projected negative profit as the USD grows stronger. Both the 50% forward + 50% option and 25% forward + 75% option strategies also has projected operating profits of 0 at some point along the exchange rate axis. This is not ideal as AIFS needs some positive liquid cashflow to tide against market downturns.

Appendix 4d: Calculation of optimal option and forward combination (crisis case)

Revenue net of non-travel costs per participant = \$1350

Cost per participant = €1000

Operating cash flow per participant = \$1350 - €1000 x $S_{USD/EUR}$

Sales volume (crisis scenario) = 10,000

Total cost (crisis scenario) = 10,000 x €1000 = €10M

Total unhedged operating cash flow = 10,000 x (\$1350 - €1000 x $S_{USD/EUR}$)

Option premium cost for base case = €25M x 5% x \$1.22/€ = \$1.525M

To find the optimal combination of options and forwards, AIFS needs a positive operating cash flow that is indifferent to the state of the world (i.e. make the line to the left of the strike price

of \$1.22 for euro flat as seen in Appendix 4c so that it does not touch zero).

Revenue in crisis scenario = 10,000 x \$1350 = \$13.5M

We assume AIFS performs its hedging strategy in 2003 for 2004 for ACIS (high school division).

Assume AIFS buys z forwards and $(1-z)$:

$$z + (1 - z) = 1 \quad (5)$$

As a base case, we assume an ex ante stable dollar (\$1.22/€ forward rate) and a projected sales volume of 25,000 (high school students) for the year 2004. As such, projected cost will be:

$$25,000 \times €1,000 = €25M \quad (6)$$

To have €25M, AIFS will have bought €25M x \$1.22/€ of forward contracts. However, in a crisis scenario, total cost is only €10M. As AIFS have overbought euros, they would need to sell it at the spot rate, denoted by s . The cost is therefore:

$$(25 \times 1.22 - 15s) \times z \quad (7)$$

We do not exercise options that are out of the money, however we have paid the option premium and will need to buy €10M at the 2004 spot rate. As such, this cost is:

$$(10s + 1.525) \times (1 - z) \quad (8)$$

As such, the profit is:

$$13.5 - (25 \times 1.22 - 15s) \times z - (10s + 1.525) \times (1 - z) \quad (9)$$

which can be simplified as:

$$11.975 - 28.975z - s(10 - 25z) \quad (10)$$

If we do not want our profit to be affected by exchange rate, we let the $s(10-25z)$ be 0:

$$(10 - 25z) = 0 \quad (11)$$

$$25z = 10 \quad (12)$$

$$z = 0.4 \quad (13)$$

$$1 - z = 0.6 \quad (14)$$

We can sub $z = 0.4$ into equation (9) to get the profit = \$0.385M which is our biggest lower bound on the profit.

Appendix 4e: Graph of optimal combination of forwards and options for ACIS department

